LEVEL 13 — Joining Arrays with NumPy

Learn np.concatenate(), np.stack(), np.vstack(), np.hstack(), np.dstack(), np.column_stack(), and np.row_stack().

1. np.concatenate()

Joins arrays along an existing axis. It does not create a new axis.

```
import numpy as np

a = np.array([1, 2, 3])
b = np.array([4, 5, 6])

print(np.concatenate((a, b)))
# Output: [1 2 3 4 5 6]
```

2. np.stack()

Joins arrays along a new axis. The input arrays must have the same shape.

```
a = np.array([1, 2, 3])
b = np.array([4, 5, 6])

print(np.stack((a, b)))
# Output:
# [[1 2 3]
#  [4 5 6]]

print(np.stack((a, b), axis=1))
# Output:
# [[1 4]
#  [2 5]
#  [3 6]]
```

3. np.vstack()

Stacks arrays vertically, usually increasing the number of rows.

```
a = np.array([[1, 2], [3, 4]])
b = np.array([[5, 6], [7, 8]])

print(np.vstack((a, b)))
# Output:
# [[1 2]
#  [3 4]
#  [5 6]
#  [7 8]]
```

4. np.hstack()

Stacks arrays horizontally, usually increasing the number of columns.

```
print(np.hstack((a, b)))
# Output:
# [[1 2 5 6]
#  [3 4 7 8]]
```

5. np.dstack()

Stacks arrays along the third axis (depth). For 2-D arrays, this creates a 3-D result.

```
a = np.array([[1, 2], [3, 4]])
b = np.array([[5, 6], [7, 8]])

print(np.dstack((a, b)))
# Output:
# [[[1 5]
#   [2 6]]
#
```

```
#  [[3 7]
#   [4 8]]]
```

6. np.column_stack()

Stacks 1-D arrays as columns. For 2-D arrays, it behaves like horizontal stacking.

```
a = np.array([1, 2, 3])
b = np.array([4, 5, 6])

print(np.column_stack((a, b)))
# Output:
# [[1 4]
#  [2 5]
#  [3 6]]
```

7. np.row_stack()

Stacks 1-D arrays as rows. It is equivalent to vstack() for this common use.

```
print(np.row_stack((a, b)))
# Output:
# [[1 2 3]
#  [4 5 6]]
```

Quick Difference: When to Use Which?

Function	Main idea	Typical result
concatenate()	Join along an existing axis	Longer existing dimension
stack()	Join along a new axis	Adds a dimension
vstack()	Stack vertically	More rows
hstack()	Stack horizontally	More columns
dstack()	Stack in depth	More depth / 3rd axis
column_stack()	1-D arrays become columns	Columns
row_stack()	1-D arrays become rows	Rows

Easy Memory Trick

concatenate = join existing axis
stack = create a new axis
vstack = vertical / rows
hstack = horizontal / columns
dstack = depth / 3rd axis
column_stack = columns
row_stack = rows

Practice Tip

Use **array.shape** before and after each operation. Watching the shape change makes the differences much easier to understand.